

**AIDS Policy and Pharmaceutical Patents:
Brazil's Strategy to Safeguard Public Health**

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Abstract

Developing nations are challenged to strike a balance between their obligations as members of the World Trade Organization (WTO) and their drug pricing strategies. The Brazilian approach to pharmaceutical price negotiations has been strikingly effective. Describing the context of the Brazilian pharmaceutical sector, their public health system, and the Brazilian AIDS policy, this paper examines the Brazilian strategy vis-à-vis the international pharmaceutical manufacturers to explore why their tactics were successful and the potential for wider application by other developing countries.

Summary:

In our paper, we examine the relationship between Brazil's obligations as a member of the World Trade Organization (WTO) and its drug pricing strategy for HIV/AIDS drugs. Brazil's strategy has been controversial though successful, insofar as they have compelled the research based pharmaceutical industry to lower pharmaceutical prices, primarily through the threat of using compulsory licenses for drugs which are legally permissible under international law. This paper examines why the Brazilian strategy was effective and considers the wider implications for other developing countries dealing with the HIV/AIDS crisis, pharmaceutical patent obligations, and the need to provide cost-effective medicines for their populations. First we provide the context for the analysis is developed through a description of the WTO's Agreement on Trade-Related Aspects of Intellectual Property Rights (TRIPS). Second, we provide a brief overview of Brazil's public health system and pharmaceutical sector. Third, we describe the Brazilian AIDS Policy. Fourth, we analyze the Government of Brazil's strategy vis-à-vis the international pharmaceutical manufacturers in terms of a game theory model and Brazil's response to the TRIPS Agreement. Finally we examine how the Brazilian case may be applicable to other developing countries in their efforts to improve their population's access to medicines. We conclude that with the precedent setting threat of a compulsory license, future discounts on patented medicines are likely. To a lesser degree, the threat of a compulsory license has already altered the *ex-ante* probabilities that the pharmaceutical firms place on such an event. Brazil's successful negotiations will thus has encouraged and will continue to encourage other developing countries to follow suit, particularly in view of the August 2003 Doha Accord.

Keywords:

Intellectual Property Rights, Pharmaceuticals, World Trade Organization (WTO), Latin America, Brazil

1 Introduction

This article examines the relationship between Brazil's obligations as a member of the World Trade Organization (WTO) and its drug pricing strategy for HIV/AIDS drugs. Brazil's strategy has been controversial though successful, insofar as they have compelled the research based pharmaceutical industry to lower pharmaceutical prices. This paper examines why the Brazilian strategy was effective and considers the wider implications for other developing countries dealing with the HIV/AIDS crisis, pharmaceutical patent obligations, and the need to provide cost-effective medicines for their populations. The paper is organized as follows. The context for the analysis is developed through a description of the WTO's Agreement on Trade-Related Aspects of Intellectual Property Rights (TRIPS) in section 2, and an examination of Brazil's public health system and pharmaceutical sector in section 3. The Brazilian AIDS Policy is considered in section 4. A game theoretic analysis of the Brazilian strategy is examined in section 5. Brazil's response to the TRIPS Agreement and its outcome are considered in section 6. The implications for other developing countries are outlined in section 7. Section 8 concludes.

2 Background of the TRIPS Agreement

The Agreement on Trade-Related Aspects of Intellectual Property Rights (TRIPS) entered into force with the formation of the World Trade Organization (WTO) in January 1995, establishing minimum standards for the protection of intellectual property rights for all signatories.¹ The Agreement covers a range of intellectual property issues beyond patents, such as trademarks, industrial designs, and copyright applicable to any sector.² It provides minimum standards for intellectual property law as well as procedures and remedies so rights holders can

¹ The text of the TRIPS Agreement may be found at www.wto.org.

² Patent protection for pharmaceuticals is exclusively examined in this paper.

effectively enforce their rights. The main patent-related component of TRIPs is that they should be available for any invention, whether product or process, in all fields of technology without discrimination. The minimum obligations for pharmaceuticals are: pharmaceutical products and microorganisms are patentable for at least twenty years from the date the inventor files for the patent application. Second, there is no discrimination permitted against patent rights for imported products. Third, exclusive marketing rights are granted until patent expiry. In May of 1996, Brazil enacted legislation to amend its patent laws, bringing them into compliance with TRIPS standards. Law 9.279 took effect May 15, 1997.³ For the pharmaceutical industry, both international and domestic, a significant aspect of the new law is the patentability of chemical products and processes, previously non-patentable.

Critics of the TRIPS Agreement argue that its implementation will lead to higher prices and further the already dire situation of limited access of the poor to medicines.⁴ The challenge for WTO member countries, particularly developing countries, is to find a balance between complying with WTO obligations and providing affordable, accessible medicines. The degree to which the mandates of the TRIPS Agreement will be felt largely depends on the pre-TRIPS patent regime (Maskus 2000). The effects will be most strongly felt by countries, such as Brazil, that have historically relied on a domestic generic industry or pursued generic substitution.

While the TRIPS Agreement does constrain production options available to developing countries, a number of the provisions can be utilized to benefit national health policy and the domestic pharmaceutical industries. These components provide signatories with limited room to

³ A description of the components of Law 9.279 that changed the Brazilian patent law is available from the Brazilian government, "INPI – The Brazilian Industrial Property System", at www.inpi.gov.br/ingles.htm.

⁴ Scherer and Watal (2001) and Maskus (2000) both survey the existing literature and note that the majority of the studies point to higher prices.

comply with the TRIPS Agreement, while operating within the boundaries of their own legal systems. Thus far, Brazil maneuvered within these confines, balancing both the obligations of the Agreement and the health and industrial interests of the nation. Table 1 describes each of the potential safeguards within the Agreement and the extent to which Brazil has utilized them.⁵ The focus of this paper is limited to the compulsory licensing allowances of the TRIPS Agreement.

Table 1

Safeguard	Brazilian Implementation
Compulsory Licensing	Yes, announced August 23, 2001 for nelfinavir, a protease inhibitor produced by Hoffman la Roche and used to treat HIV. Unnecessary following further negotiation with the company. The law makes provisions for compulsory licensing under the following circumstances: dependent patents, national emergencies, public interest, and abuses of patent rights
Protection of Public Health	Yes, utilized for HIV/AIDS drugs.
Transition Period	No. The Agreement allowed Brazil until 2005 to bring its laws into TRIPS-compliance. Brazil complied ahead of schedule when Law 9.279 took effect on May 15, 1997.
Pipeline Protection	No. While TRIPS does not require such protection, Brazil granted pipeline protection for pharmaceuticals in September of 1996.
Institution of Minimum Standards	No.
Selective Patentability	Yes. The 1996 Brazilian law excludes the patentability of 'biological materials found in nature'. This also includes genome and germplasm of living beings.
Early Working (Bolar) Exception	No.
Local Working Requirement	Yes. According to article 68 of the Brazilian Patent Law, patent rights are contingent on firm's setting up manufacturing facilities in Brazil and are otherwise subject to compulsory licensing.

(Lybecker 2001)

⁵ A complete description of each of these provisions is included in Appendix 1. Countries are only obligated to create laws that institute the minimum standards required by TRIPS. Adherence to strictly the minimum standards does safeguard against limited access to medicines and higher pharmaceutical prices.

3 Brazil's Public Health System and Pharmaceutical Sector

The Brazilian health system has both public and private facilities ranging from basic health care units to complex hospital facilities (Brazilian Ministry of Health 2000a). Pursuant to Article 196 of the 1988 Constitution, access to health services, including access to basic medicines, is considered a constitutional right in Brazil. The Constitution expresses, “Health is a right of all and a duty of the State and guaranteed by means of social and economic policies aimed at reducing the risk of illness and other hazards and all the universal and equal access to actions and services for its promotion, protection and recovery”.⁶ Public health care service delivery is shared equally by the different levels of government: federal, state, municipal, and the national government health system (*Sistema Unico da Saude* [SUS]). In practice, the delivery and management of health services is increasingly being decentralized to the state and municipal levels, reflecting the government’s sensitivity to the population’s preference for more local governance. The federal level of government defines the policies and regulations, grants technical and financial support to the states and municipal governments and provides some service delivery. These governments, in turn, contribute the remainder of the health budget and share responsibility for health service delivery.

Brazil has embarked on a process of health system decentralization, primarily to respond to the inefficiencies of the previous centralized administration of the military regime. The decentralization of health care system delivery has largely been implemented through the Basic Care Floor (*Piso de Atencao Basico* [PAB]) and the Family Health Program (*Programa de Saude da Familia* [PSF]). The PAB guarantees the financing of select ambulatory services for municipalities capable of managing them. The federal government transfers resources to the

⁶ Article 196 of the 1988 Brazilian Constitution.

municipal governments on a per capita basis, providing the majority of the financial resources for public health in Brazil, about 71%. States contribute about 15% and municipalities the remaining 14% (World Bank 1999b). States and municipalities also have the right to specific tax and expenditure functions, and are entitled to take over full management of basic health care for their respective health systems. They also can choose to “opt-out” of managing health services, and let the federal government take responsibility for their provision. This is usually the case for poorer municipalities and/or states, particularly in the supply of basic medicines.

Although Brazil’s domestic pharmaceutical industry has a small share of the global pharmaceutical market, it is a major supplier to the domestic public health system and has historically benefited from decades of a regime of non-patent protection.⁷ This has contributed to the domestic industry’s dominance of the Brazilian market, as shown in Table 2. The legal right of local firms to produce on-patent drugs was expressed in Law No. 5772 on Industrial Policy that took effect on December 21, 1971.⁸ As a result, during the 1970s many private firms, such as Ache, Farmasa, Libbs, Sintofarma and public producers, such as Fiocruz in Rio de Janeiro and FURP in Sao Paulo, benefited from legally reverse-engineering on-patent pharmaceuticals.

Table 2

Brazilian Market Statistics (US \$ billion)				
	1997	1998	1999*	2000*
Total Market Size	12.7	10.3	11.0	11.5

⁷ The Brazilian market has shrunk considerably lately, but remains the 10th largest pharmaceutical market. In 2002, the global pharmaceutical industry had sales of \$406 billion (IMS Health 2002).

⁸ In this context, “on-patent drugs” refers to drugs that are protected by patents in markets where such patents are available to pharmaceutical products, usually the United States or Europe rather than in Brazil.

Local Production	11.6	9.5	10.0	10.3
Imports	1.2	1.0	1.3	1.5
Exports	0.1	0.2	0.3	0.3
Imports from U.S.	0.2	0.2	0.2	0.2

(Mendonca 1999) Data from SECEX (Brazilian Government Statistics) and ABIFARMA.

* estimate

Pursuant to the TRIPS Agreement, the Industrial Properties Law was signed by President Cardoso on May 14, 1996 (number 9279) and it came into force on May 15, 1997. The Law provides a higher level of protection for pharmaceutical patents than that found in most other developing countries. The law allows for the patentability of chemical products, a change that critics feared might result in higher pharmaceutical prices. In response, Brazil passed a Generic Drug Law⁹, facilitating generic drug entry into the marketplace and requiring that under the government system all purchasing contracts are to be generic drugs and all prescriptions are to be written using generic names.¹⁰ This is significant because the Brazilian public health system represents 25% of the pharmaceutical market.¹¹ Equally important, the strategy of purchasing generic drugs will likely benefit local producers and help contain pharmaceutical expenditures.

Despite the public provision of medicines in health care facilities in Brazil, as in other developing countries, the majority of pharmaceutical expenses constitute out-of-pocket expenditure as a result of limited efficacy of public pharmaceutical systems (WHO 1998). The Brazilian Ministry of Health estimates that as much as 50% of the population does not have regular access to basic medicines. An additional 20% of the population has partial coverage, and

⁹ It was passed in November 1998 in the House of Representatives and Senate approval followed in January 1999.

¹⁰ Generic names were mandated on all packaging, all prescriptions were to be identified by their generic names, and use of generic drugs was legislated in all government purchasing proposals and contracts (SCRIP 1999b).

¹¹ The market for generics is expected to increase by 40% over the next few years, saving the government \$400 million annually and reducing pharmaceutical prices for consumers by 40-45% (Mendonca 1999).

only 30% of the population is able to satisfactorily meet their pharmaceutical needs given the inability of the public health system to guarantee the supply of essential medicines for the poor (Brazilian Ministry of Health 2000b).

Notwithstanding the constitutional commitment to universal health care, the population's access to health services and the quality of such services varies according to geography, income, and government resources. The SUS contracts out a large majority of inpatient and outpatient care to a network of private and philanthropic hospitals, clinics and other facilities. While SUS is responsible for providing and financing health care for all of the population, its clients are mainly the 123 million Brazilians (74%) who cannot afford to be covered by the private sector (World Bank 1999b). The differences in access to medicines in Brazil are profiled in Table 3, across income groups. ABIFARMA (*Associação Brasileira da Indústria Farmacêutica*), the Brazilian pharmaceutical industry association, compiled this information describing the tiered nature of the pharmaceutical market. Consumers are classified into three income groups: A, B, and C. Consumers in Group A comprise 15% of the population and consume 48% of all pharmaceuticals sold in Brazil. Group B is made up of 34% of the population and these consumers purchase 36% of all pharmaceuticals sold. Lastly, Group C is made up of the poorest 51% of the population, consuming only 16% of the pharmaceuticals sold in Brazil.

Table 3

Profile of the Brazilian Pharmaceutical Consumer				
Group	Salary Range (monthly basis)	Percentage of the Population	Percent Market Share Consumed	Per Capita Expenditure on Pharmaceuticals
A	10 + minimum salaries	15	48	R\$205
B	4-10 minimum salaries	34	36	R\$ 68
C	0-4 minimum salaries	51	16	R\$ 20

(Cohen 2000) Source: ABIFARMA 1998. A minimum salary is approximately US\$100 per month. In November 1998, R\$1.19=US\$1.00.

Pharmaceutical consumption is also unevenly distributed across geographic regions. In a 1998 study of the pharmaceutical industry of Brazil, the World Bank examined the problematic structure of the sector, noting that the “severe under financing of the public system results in regional inequalities, arbitrary rationing of facilities, and a perceived decline in quality” (World Bank 1998). A more recent study echoed these conclusions, appraising the scarcity of basic medicines and health care services, and tying regional inequalities to differences in urban development rather than income. The more rural regions have more limited access to pharmaceuticals and other health services (World Bank 1999b).

4 AIDS Policy in Brazil

A principle reason behind the Brazilian government’s strong commitment to the HIV/AIDS program, particularly with regards to the free and widespread distribution of antiretrovirals, is the effective, well-organized lobbying from civil society (Galvao 2002).¹² The government’s constitutional commitment to universal health care for the population has been interpreted to include the provision of medicines used for the treatment of HIV/AIDS and has been made feasible by the capacity of the government to turn to local producers for the production of the majority of medicines required to treat HIV/AIDS. Still, this commitment has consumed a large share of the Federal drug budget.

Table 4

Total Cost of Anti-retroviral Therapy (\$US million)	
Brazil 1997-2000	
Year	Total Cost (\$US million)

¹² Representative groups such as individuals living with HIV/AIDS; gays and lesbians; feminists; and faith-based groups helped pressure the government to respond effectively to the HIV/AIDS crisis (Galvao 2002).

1997	174
1998	241
1999	281
2000	258
Federal Allocations for Drug Procurement	
1999	552
2000	521

(Bermudez 2002, Szwarcwald 2002)

As previously noted, the nonrestrictive nature of Brazilian patent law allowed domestic producers to manufacture their own nucleoside analogues, beginning in the mid-1990s. Domestic production has been an important component of the Brazilian policy and continues to be. Pursuant to the TRIPS Agreement, Brazil has invoked the “national emergency” clause to facilitate local production of antiretrovirals.¹³ While significant, “47% of antiretrovirals (19% of expenditure) were acquired from domestic firms . . . 53% of antiretrovirals (81% of expenditure) were acquired from international pharmaceutical companies” (Galvao 2002). Nevertheless, the approach has been successful in managing drug expenditures as well as confronting the AIDS crisis. Between 1997 and 2001, the Brazilian Ministry of Health estimates cost savings of \$1.1 billion (Galvao 2002). Continued success hinges on the ability of the government to provide affordable drugs to HIV/AIDS patients. Moreover, a critical factor in the affordability of these treatments may be domestic production, or at least the credible threat of domestic production.

¹³ As regards other use without authorization of the rights’ holder, Article 31(b) of the TRIPS Agreement states that, “the following provisions shall be respected . . . such use may only be permitted if, prior to such use, the proposed user has made efforts to obtain authorization from the right holder on reasonable commercial terms and conditions and that such efforts have not been successful within a reasonable period of time. This requirement may be waived by a Member in the case of a national emergency or other circumstances of extreme urgency or in cases of public non-commercial use. In situations of national emergency or other circumstances of extreme urgency, the right holder shall, nevertheless, be notified as soon as reasonably practicable.”

5 A Game Theoretic Analysis of the Brazilian Strategy

Brazil has utilized the flexibility of many of the provisions of the TRIPS Agreement, interpreting them so as to adjust the imperatives of the Agreement to the domestic realities. The focus of this article is limited to the issues surrounding compulsory licensing of HIV/AIDS drugs.¹⁴ Generalizing the strategic interactions is most easily done through the game theoretic model described in Figure 1 below. This model depicts the negotiations between Brazil and Hoffmann-La Roche Inc. (Roche), but more generally describes, albeit at an elementary level, the interaction of any representative developing country and pharmaceutical firm. Wider application of the model is described at more length in the implications discussion that follows.

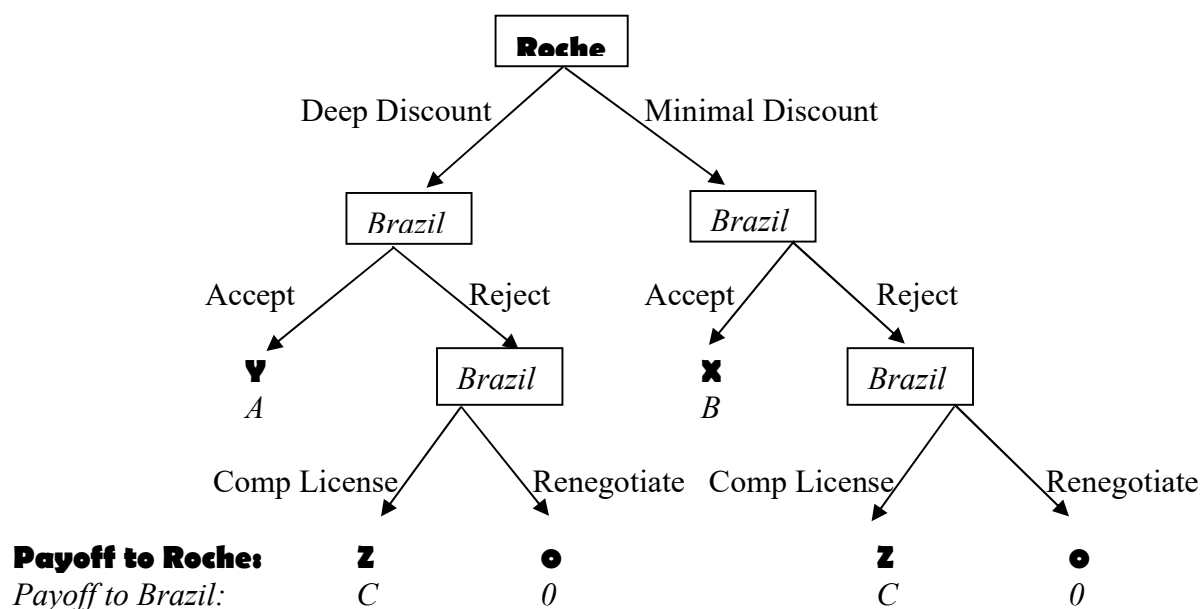
The simplified version of the game features two players, Brazil and Roche. Each player has a choice of strategies. Roche moves first, offering a discounted price for the drug under negotiation, selecting either a deep discount or a minimal discount. In making its offer, Roche recognizes that Brazil has three options (accept offer, reject offer and issue a compulsory license, reject offer and renegotiate).¹⁵ Roche has some *ex-ante* expectation of the probability of each of Brazil's options, given the degree to which they discount the drug's price. Clearly, a deeper

¹⁴ A compulsory license is a license to produce granted by government authorities to a third party to "make, use or sell the patented product for a fixed time period during the life of the patent, even without the consent of the patent owner, upon the payment of a reasonable remuneration" (Watal 2000).

¹⁵ Obviously the firm may choose not to sell to the country and the country may choose not to purchase from the firm. These strategies are ignored here.

discount will increase the likelihood that Brazil will accept the offer; while a minimal discount increases the likelihood Brazil will issue a compulsory license for the drug under negotiation.¹⁶

Figure 1



Following the offer from Roche, Brazil decides whether to accept the offer, reject and issue a compulsory license, or reject and renegotiate. In the first two cases, the game ends. In case of the latter, the game is repeated. The possible outcomes may be described as follows.

Outcome 1: if Roche offers a deep discount and Brazil accepts the offer, Roche produces the drug and Brazil purchases it at the agreed upon price. The payoff to Brazil is A , and to Roche, Y .

Outcome 2: if Roche offers a minimal discount and Brazil accepts the offer, Roche produces the drug and Brazil purchases it at the agreed upon price. The payoff to Brazil is B , and to Roche, X .

¹⁶ Note that if Brazil rejects the offer of a minimal discount, the choice of whether to issue a compulsory license or renegotiate remains. If the probability of renegotiation is higher following an offer of a minimal discount, then such an offer does not necessarily mean that a compulsory license is more likely.

Outcome 3: if Brazil rejects the offer and issues a compulsory license, Brazil begins production of the drug under patent. Regardless of the discount offered, if Brazil issues a compulsory license, Roche receives payoff Z , while Brazil receives payoff C . [Note that it is possible that Roche could choose to counter with a new offer and the game would repeat.]

Outcome 4: if Brazil rejects the offer and renegotiates, payoffs to each are zero. Roche updates its *ex-ante* expectation and again chooses whether to offer a deep discount or a minimal discount and the game repeats.

Certainly the pharmaceutical firm prefers to offer a minimal discount over the deep discount, but the deep discount is preferred to losing the developing country market to a compulsory license. That is, $X > Y > Z$. As for the developing country, they prefer the deeply discounted drugs to a minimal discount. It follows that $A > B$. Alternatively, the country may decide to issue a compulsory license. It is reasonable to assume that a minimal discount, results in higher drug prices than under domestic production. Accordingly, $C > B$. A comparison of A and C is more difficult. Depending upon the extent of the (deep) discount and the costs of domestic production, A may be greater than, equal to, or less than C .

The strategic interactions between the Brazilian government and Roche played out as follows. Rejecting the initial Roche offer, a minimal discount deemed to be too low, on 22nd August 2001, the then Minister of Health, Jose Serra, announced Brazil had started the process to domestically produce nelfinavir (marketed as Viracept). The Minister began the process of issuing a compulsory license, invoking an article in the Brazilian Constitution that allows for

such licensing in the case of abusive prices (Rich 2001). With the Doha talks fast approaching, Brazil was on the threshold of the first infringement of an HIV/AIDS treatment drug patent.¹⁷

Given that Brazil possesses the manufacturing capacity to produce the drug, the threatened compulsory license was credible and significant. The Brazilian pharmaceutical market is an important one, one of the largest in Latin American and one of the top ten globally.¹⁸ Both of these factors reduce the size of Z, the payoff to Roche under a compulsory license. In response to the Minister's announcement and the looming issuance of a compulsory license, Roche offered to lower the price of nelfinavir by 13%. This offer was also rejected by Brazil as unacceptable and preparations were made for Far-Manguinhos, a state-owned laboratory and a part of FIOCRUZ, to produce the drug. Roche then offered to cut the price of nelfinavir by one-third, following the Brazilian request for a discount of 40% (Jordan 2001).¹⁹

¹⁷ This infringement is legal under Brazilian law given a "national emergency". While viewed as vital in the face of the AIDS crisis, it also important to note that Minister Serra knew that his threat to resort to compulsory licensing would be widely supported by the population and help him gain political capital for the 2002 Presidential campaign.

¹⁸ The Brazilian economy is the 8th largest in the world. Demographic and epidemiological changes have increased the Brazilian demand for pharmaceuticals, particularly those for chronic diseases such as diabetes and tuberculosis. Leading up to the recent financial crisis Brazil was the largest pharmaceutical market in Latin America. The Mexican market is now somewhat larger. In 1998 Brazil was the world's 6th largest market for pharmaceuticals, with three percent of worldwide sales (Lybecker 2000).

¹⁹ Again, this offer was not thought to be low enough. At a cost of over \$88 million for Roche's Viracept, this single drug accounted for 28% of government spending on antiretrovirals in 2000. The Brazilian government spent \$303 million in 2000 to provide the HIV/AIDS cocktail to approximately 100,000 AIDS patients (Rich 2001).

As the Roche negotiations continued, the Brazilian Health Ministry continued securing lower prices for many of the other HIV/AIDS cocktail drugs.²⁰ Merck acquiesced to the pressure, offering deep discounts of 65% and 59% on the two HIV/AIDS drugs they provide to Brazil (Rich 2001). As for the remaining drugs, Brazil was producing eight of the twelve drugs utilized in the HIV/AIDS cocktails. This was done without violating patents given that the drugs were being manufactured prior to the passage of the new intellectual property law and thus fell outside the exigencies of the TRIPS Agreement. The Brazilian approach consisted of importing generic raw materials and utilizing public manufacturing facilities, such as FURP and FIOCRUZ (via Far-Manguinhos), to locally manufacture the drugs.²¹

6 The Outcome and its Determinants

While outcome of this game is obviously dependent upon the size of the payoffs for both Brazil and Roche, the factors that determine the sizes of these payoffs are less apparent. Consider first the payoffs to Brazil. A deep discount or issuance of a compulsory license are preferred to a minimal discount. In the case of either a deep or minimal discount, the payoff to Brazil is quite straightforward: the negotiated cost savings on the required number of doses.

The Brazilian payoff from the issuance of a compulsory license is more complex. The short-run and long-run implications of the license must be taken into account. In the short-run,

²⁰ Currently the Ministry of Health makes 12 antiretrovirals available: Zidovudine, Didanosine, Zalcitabine, Lamivudine, Stavudine, Indinavir, Ritonavir, Saquinavir, Nelfinavir, Nevirapine, Delavirdine, and Efavirenz (Brazilian Ministry of Health 2002). In accordance with patient needs, treatment consists of three or four drugs.

²¹ The share of raw materials imported for manufacture varies. In 1999, Brazil imported as much as 70% of its raw materials (SCRIP 1999a). As of 1999, domestic Brazilian firms were supplying 47% of antiretrovirals (19% of expenditure), of which 92.5% were state-run facilities and 7.5% were private enterprises (Galvao 2002).

the Brazilian government must consider the time and expense of domestic production. While the Brazilian industry presumably has sufficient domestic capacity, as well as established technical experience, it is worth noting that domestic production is heavily subsidized in Brazil.²² The extent to which this would continue, and for how long, must enter the government's calculations.

Beyond available capacity and production costs, several other factors must be taken into account in estimating the payoffs to Brazil. First, in the market for pharmaceuticals the first-mover-advantage is a significant factor in determining market share. To fully capture the welfare benefits of such a license, it is important that licensees enter early (Watal 2000). For domestic manufacturers the ability to gain market share, foster competition and lower prices, hinges on early entrance. In addition, one must account for the royalty payment, to be made to the patent holder, as well as the administrative costs of compulsory licenses.²³ The degree to which a compulsory license will lead to competition and lower prices will critically depend on the criteria utilized to determine the amount of compensation paid to the patent holder. In an examination of the TRIPS Agreement compulsory licensing provisions, Scherer and Watal find that the "choices made in industrialized nations provide ample precedent for royalty-setting on the modest side of the range of possibilities" (Scherer and Watal 2001).

The compulsory licensing payoff for Brazil also depends on several long-run factors. To start, does the domestic industry have sufficient capacity to dependably meet demand, perhaps increasing demand, over time? Moreover, does the industry have a reliable source for the active

²² Historically, Brazil has provided annual subsidies to these firms at the taxpayers' expense (IFPMA 2002).

²³ Though not trivial, the administrative costs of compulsory licenses may arguably be lower than those of other mechanisms used to curtail high pharmaceutical prices. In a comparison of different mechanisms, Watal finds that comparable levels of welfare gains can be achieved through price controls and compulsory licenses, with lower administrative costs associated with the compulsory license (Watal 2000).

pharmaceutical ingredients (APIs) and excipients?²⁴ The feasibility of the Brazilian compulsory license strategy is contingent upon affirmative answers to both of these questions. Presumably, domestic capacity could be adjusted, as the preparations at Far-Manguinhos indicate.²⁵

Two additional long-run factors are worth mentioning. First, domestic production may limit the variety of medical interventions offered since the entire range of products may not be available through local producers. Brazilian producers primarily supply only one class of drugs, Nucleoside Reverse Transcriptase Inhibitors, to the government (IFPMA 2002). Finally, the compulsory license payoff to Brazil is also contingent on the experience gained through local production, so called learning-by-doing. The Brazilian pharmaceutical industry possesses a sufficient degree of sophistication to benefit from such experience, further cultivating an innovative domestic industry. The long-run benefits to advancing domestic innovation, though uncertain and difficult to quantify, should be weighed in calculating the payoff to Brazil.

An examination of the payoffs to Roche is somewhat more complicated. As in the case of the payoffs to Brazil, the firm must weigh all of the potential short-run and long-run consequences. On the surface, the minimal discount will ensure greater profitability for the firm

²⁴ “APIs include any substance or mixture of substances that are intended to proved pharmacological effect.

Excipients, on the other hand, are inactive ingredients and can include fillers, bulking agents, binders, disintegrants, coatings, colorants, slip agents, etc. A pill can contain just one, or more than 20, active ingredients (APIs) and inactive ingredients” (GlobalOptions Inc. 2003).

²⁵ The availability of APIs will depend upon Brazil’s ability to continue to import them from abroad. The largest producers of bulk pharmaceutical ingredients are China and India and even pharmaceutical firms in the United States import about 80% of the bulk APIs used in production (GlobalOptions Inc 2003). While Brazil’s ability to import APIs should not be impacted by issuing a compulsory license, this possibility cannot be ruled out. Given the importance of multinational pharmaceutical firms, one can imagine a situation in which such firms might successfully pressure the producers of bulk ingredients to restrict their dealings with Brazil.

than either the deep discount or the issuance of a compulsory license. However, this strategy may antagonize Brazil, increasing the likelihood of the issuance of a compulsory license. A compulsory license would mean the loss of significant market share in Brazil, though the firm would receive a royalty. While royalty rates vary greatly, from estimates of 18% in the U.K. to 4% in Canada, the firm would prefer not to license the drug (Watal 2000).²⁶ More important, the license would generate a great deal of media attention, pointing to the firm's inability to reach an agreement with the Brazilian government and the perception of high drug prices. In addition, the firm must bear in mind the implications of the precedent set by a compulsory license. It is reasonable to assume that the possibility of other developing nations issuing compulsory licenses increases following the first issuance. The firm must consider an increased future probability of additional licenses and the associated losses from those licenses.

In like manner, the firm must consider the consequences of a deep discount. The firm's preference between a deep discount and the compulsory license will depend on several factors. Primary among these is the extent of the discount. While a deep discount should preserve the firm's market share in Brazil, it does reduce the firm's profits in Brazil. This strategy is subject to another risk for the firm. Specifically, the firm risks that the deep discount will be used against them in the context of external reference pricing in other markets.²⁷

Ultimately, Brazil's threat to manufacture patented drugs unless the pharmaceutical companies negotiated for lower prices resulted in further negotiations and a settlement for

²⁶ Watal (2000) cites a 1979 study by Temin pointing to the reluctance of drug firms to license their patents.

²⁷ External reference pricing refers to the practice by some governments to consider the lowest prices offered in other markets in determining the acceptable level in the domestic market.

continued production by the multinational firms. Their strategy worked for a number of reasons. For one, Brazil had the capacity to manufacture the drugs so the threat was perceived as credible. Secondly, Brazil is a large and lucrative pharmaceutical market, so the pharmaceutical industry stood to lose an important market if Brazil went forward with the compulsory licensing of patented medicines. The Brazilian case also followed on the heels of the landmark case filed against the Government of South Africa by a Pharmaceutical Industry Association and 39 affiliate companies, addressing its Medicine and Related Substances Control Amendment Act.²⁸ The awareness raised by this case certainly strengthened the Brazilian position. In addition, the Brazilian position was strongly supported by local advocacy groups and the international health community.²⁹ The research based pharmaceutical industry, particularly after the passage of the TRIPS Agreement, has been the focus of an international campaign led by a network of health activists, including nongovernmental organizations, such as Oxfam and Médecins Sans Frontières/Doctors without Borders, to ensure access of the poorest members of the global population to medicines. This health network has focused much attention on the provision of the TRIPS Agreement and aggressively critiqued it as an unfair imposition on developing countries. The network has applied constant pressure to the multinational pharmaceutical firms for lower prices. Their constant chants that firms place “profits over people” echoed in the international media. This undeniably assisted the Brazilian government in securing lower drug prices.

7 Implications for Other Developing Countries

²⁸ Ultimately, the resolve of the South African government and international public pressure resulted in the withdrawal of the companies from the case (WTO 2001b).

²⁹ While the focus of this work isn't the international campaign, the cumulative impact of this pressure through a multitude of channels – scholars, press, other governments – is undeniable.

The degree to which the Brazilian negotiations will translate into real change and enhanced bargaining power for other developing nations hinges on several factors. Most important among these is the resolution of the questions surrounding TRIPS allowances of compulsory licenses. While still under debate, the existing interpretation of the TRIPS Agreement allows a country to issue a compulsory license for a drug, which is then produced for them elsewhere.³⁰ The interpretation is particularly important to those developing countries that do not possess the capacity to produce drugs domestically, particularly least developing countries. For example, Brazil would be able to export a drug to Zambia after Zambia issued a compulsory license for it. While this interpretation provides a safeguard for the provision of critical medicines, there is currently pressure from the United States Trade Representative Robert Zoellick for an interpretation of the trade agreement that would only permit drugs under a compulsory license to be produced domestically within the nation issuing the license (Health GAP Coalition 2001).

The importance of this issue is perhaps best illustrated in the 2001 Doha Declaration on TRIPS and Public Health.³¹ In part, the Doha talks sought to clarify article 31(f) of the TRIPS Agreement that states that compulsory licensing shall be “predominantly for the supply of the domestic market”. Leading up to the Doha talks, developing nations sought an expansive

³⁰ The TRIPS requirement states “any compulsory drug patent license be authorized predominantly for the supply of the domestic market . . . The ‘predominantly’ term in Article 31(f) clearly implies that some exportation under compulsory license in the exporting nation will be allowed” (Scherer and Watal 2001).

³¹ Available at http://www.wto.org/English/thewto_e/minist_e/min01_e/mindecl_trips_e.htm.

interpretation³² of Article 31(f) and requested language in the ministerial declaration providing that ““nothing in the TRIPS Agreement prevents Members from granting compulsory licenses for foreign suppliers to provide medicines in the domestic market’ and ‘nothing in the TRIPS Agreement will prevent Members to grant compulsory licenses to supply foreign markets”” (Sykes 2002). How the debate is resolved is crucial to ensuring access to medicine for the host of nations that rely on imports. As noted by Scherer and Watal, UNIDO statistics indicate that “46 of the 133 non-OECD countries . . . imported 100 percent of their requirement of medicines in 1989. This count goes up to about two-thirds when we include all nations importing more than 50 percent of consumed medicines by value” (Scherer and Watal 2001).

Ultimately, the issue remains unresolved and the Doha Declaration instructed the Council for TRIPS to look for a solution. However, the declaration did include the provision that “...the TRIPS Agreement does not and should not prevent members from taking measures to protect public health. We affirm that the Agreement can and should be interpreted and implemented in a manner supportive of WTO Members' right to protect public health and, in particular, to promote access to medicines for all.” Clearly the ultimate interpretation of the law will have far-reaching consequences in the global market for HIV/AIDS drugs. This is particularly true given their high cost and demand, but will also greatly impact drugs used to treat other diseases and conditions. The success of the Brazilian program points to the importance of the compulsory licenses and the essential nature of its availability for all developing countries in confronting an increasingly complicated and costly global disease burden.

³² “Members should take the view that the TRIPS Agreement in no way stands in the way of public health protection, and therefore it should provide the **broadest flexibility** for the use of compulsory licenses” (WTO 2001b, emphasis added).

The Brazilian strategy has clearly altered the negotiating environment between developing countries and innovative pharmaceutical firms. While the issuance of a compulsory license was always a possibility, the invocation of such a license increases the probability that it will be used in the future by setting a precedent.³³ Increasing the probability of the use of compulsory licenses may affect the pricing strategies of multinational pharmaceutical firms. Specifically, given the ability to threaten domestic production, the discounted prices granted to developing countries should drop to the level of domestic manufacture under compulsory license.³⁴

Beyond increasing the probability of the issuance of a compulsory license, the Brazilian negotiations have contributed to a gradual, though unmistakable, cumulative change in HIV/AIDS policymaking and pharmaceutical negotiations.³⁵ Consider the recent changes taking place in several developing countries. Specifically, several countries are now making preparations for domestic manufacture of antiretrovirals. Namibia, Uganda, and Ethiopia are moving forward with plans to produce generic copies of HIV/AIDS drugs (Maletsky 2003, AllAfrica.com 2003). In addition, the World Health Organization adopted the Brazilian model for combating AIDS. In 2003, the incoming WHO director-elect Jong Wook Lee asked that the

³³ In like manner, the Brazilian negotiations have increased the probability that the threat of a compulsory license will be used again in future negotiations.

³⁴ Dropping their prices to the level of domestic production allows pharmaceutical firms to remove the financial incentive for local production. Clearly the determination of price is dependent on many factors. On the one hand, large multinational firms benefit from economies of scale, lowering their production costs. At the same time, factors such as brand loyalty and reputation may keep prices above the level of domestic production costs.

³⁵ The importance of the South African case cannot be overstated in this context. The Brazilian negotiations, and other changes, have contributed to the groundswell that is shaping the environment for pharmaceutical negotiations.

(then current) head of the Brazilian Health Ministry's AIDS program, Paulo Teixeira, be released to formulate new policy based on the Brazilian experience (PhRMA 2003). Furthermore, this is not the only adoption of the Brazilian program. Teixeira noted that in the past three years, "31 developing countries have adopted Brazil's guidelines and both the Kenyan and South African governments are working with medicines donated by Brazil" (PhRMA 2003).

As evidence of additional change, Mexico is considering integrating compulsory licenses into its patent law. A bill currently pending in the Mexican Congress would amend Article 77 of the Mexican Patent Law empowering a government agency, the General Health Council, to issue compulsory licenses for drugs for "serious illnesses". The patent holder would receive a royalty up to 20 percent of the retail price. Proponents of the bill claim that prices for AIDS drugs could fall to 10 percent of their current levels (Case & Iliff 2003). The pending bill is already having an effect on the industry, such that "large drug companies have begun taking out ads in newspapers trumpeting their research capabilities and recent price cuts" (Case & Iliff 2003).

The cumulative effect is impacting the pharmaceutical industry in other ways as well. Most significantly, the German manufacturer Boehringer Ingelheim has issued an official statement that it will not interfere with the issuance of compulsory licenses.³⁶ The statement reads, "We agree with the Doha Declaration on the protection of public health and will not take any measures against countries which use the protective measures of the TRIPS Agreement or the Doha Declaration to protect their population (e.g.: compulsory licenses) respectively implement them in their national laws" (Shen 2003).

Finally, beyond the uncertainty surrounding the interpretation of Article 31(f) and the potential benefits of domestic production, developing nations are constrained by the inherent

³⁶ Boehringer Ingelheim produces Nevirapine and was one of the companies involved in the South African lawsuit.

limitations of compulsory licenses. Scherer and Watal (2001) describe several important considerations: (i) licensees must have the ability to reverse-engineer the drug, (ii) compulsory licensed products destined for export are only relevant when similar disease patterns affect the exporting and importing nations, (iii) “compulsory licensees will be only attracted to large and profitable drug markets, and so essential medicines with small potential volumes or mostly poor patients will not attract many applicants, however important it is from the perspective of public health” (Scherer and Watal 2001), and (iv) one must consider how much compensation is appropriate and the amount of royalty to return to the patent owner. Even with the resolution of the legal issues surrounding Article 31(f), developing countries still face significant challenges in bringing affordable HIV/AIDS drugs to their populations. Nevertheless, it is hoped that the August 2003 DOHA Agreement has made this policy option more viable for most developing countries and particularly for those who do not have sufficient capacity to manufacture locally but can turn to other countries, such as Canada and Brazil, for the import of medicines.

8 Conclusions

This analysis of the Brazilian strategy vis-à-vis the international research based pharmaceutical industry demonstrates how it resulted in lower prices for the patented medicines needed for the treatment of HIV/AIDS. The Brazilian case is significant insofar as what it represents for other developing countries that are in need of affordable medicines for their populations. When Brazil embarked on the issuance of a compulsory license for nelfinavir the issue was raised. One may argue that the possibility of other developing countries issuing compulsory licenses increases following the first issuance. Such an event would alter the probabilities that pharmaceutical firms place on such a strategy. With a precedent setting compulsory license, the game would be forever changed and it is likely that this would guide

future discounts on patented medicines. However, to a lesser degree, the threat of a compulsory license has already altered the *ex-ante* probabilities that the pharmaceutical firms place on such an event. Brazil's successful negotiations will encourage other nations to follow suit. As of January 1995, pharmaceutical products were excluded from patentability by 18 WTO member countries.³⁷ As these nations bring their laws into TRIPS-compliance, each of them may find itself in a similar position, facing the same choices as Brazil.

³⁷ According to Scherer and Watal (2001) the list included Angola, Argentina, Bangladesh, Brazil, Cuba, Egypt, Guatemala, India, Kuwait, Madagascar, Morocco, Pakistan, Paraguay, Qatar, Tunisia, Turkey, United Arab Emirates, and Uruguay.

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Appendix 1

Principle of Exhaustion of Rights: Following the principle of exhaustion of rights, member countries would be allowed to pursue parallel imports of pharmaceuticals. That is, the legal importation of a patent-protected product without authorization from the patent holder after the product in question has legitimately been put on the market in a second member country. In situations in which large price differentials exist between countries, parallel importing allows developing countries to “shop around”. At the same time, critics charge that the result will be uniformly high prices across all countries. They argue that the implementation and enforcement of the TRIPS Agreement will result in the marketing of patented drugs by multinational corporations in all developing countries and the maintenance of high prices by the same corporations.

Compulsory Licensing: While subject to a number of restrictive conditions, compulsory licensing is not precluded by the Agreement. Advocates contend that developing countries must incorporate compulsory licensing directly into domestic legislation to act as an effective deterrent to monopoly practices and facilitate access to licenses and a transfer of technology. This will require that they spell out the meaning of elements such as ‘reasonable commercial terms’ and ‘adequate remuneration’ in terms that are acceptable to patent holders.

Protection of Public Health: In addition, member countries may be able to use Article 8 of the Agreement to achieve some of their public health objectives. Article 8 grants Members the ability to “adopt measures necessary to protect public health and nutrition, and to promote the

public interest in sectors of vital importance to their socio-economic and technological development, provided that such measures are consistent with the provisions of this Agreement” (WTO 2001a).

Transition Periods: From its inception, the TRIPS Agreement recognized that developing countries and transition economies would need longer periods of time to come into compliance with the provisions of the Agreement. Pursuant to a recent extension, least developed countries have until 1 January 2016 to come into compliance. Countries may utilize the transition period to fully explore the flexibility of the Agreement and gradually apply the provisions to provide product and process patents to pharmaceuticals.

Pipeline Protection: The TRIPS Agreement does not require “pipeline” protection. That is, there is no requirement for retroactive protection for innovations that would be covered under the newly proposed laws, but were not eligible for protection under earlier legislation. Countries may take advantage of this by refusing to grant such protection.

Bolar Exception: The ‘Bolar Exception’ or early working provision, grants generic producers the right to test their product and secure approval from regulatory authorities prior to the expiration of the patent. This allows for the provision of lower priced generic drugs immediately upon expiry. As is the case with parallel imports and compulsory licenses, the ‘Bolar Exception’ must be written into national law. There is nothing automatic about these provisions.